Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Course Code
ECE 3112

Course Title
Microprocessor, Assembly Language & Interfacing Sessional

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Lab Report 7: Using Assembly Language to Program 8086 Microprocessor to Solve Problems Related To Array.

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Problem 7.1

Problem Description

Reverse an array.

Tools Used

- emu8086 - 8086 Microprocessor Emulator
- MacTeX -LaTeXcompiler.
- VS Code with LaTeXworkshop extension as a text editor.

Process

Code:

```
;-------------------------------------------------

DATA SEGMENT
    ARR DB 1, 2, 3, 4, 5, 6, 7, 8
DATA ENDS

CODE SEGMENT
    ASSUME DS:DATA, CS:CODE

START:
    MOV AX, DATA
    MOV DS, AX

    MOV CX, 4
    XOR AX, AX

    LEA SI, ARR
    MOV DI, SI
    ADD DI, 7

REVERSE:
    MOV AL, [SI]
    XCHG AL, [DI]
    XCHG [SI], AL

    INC SI
```

```
25      DEC DI
26
27      LOOP REVERSE
28
29  CODE ENDS
30  END START
```

Output

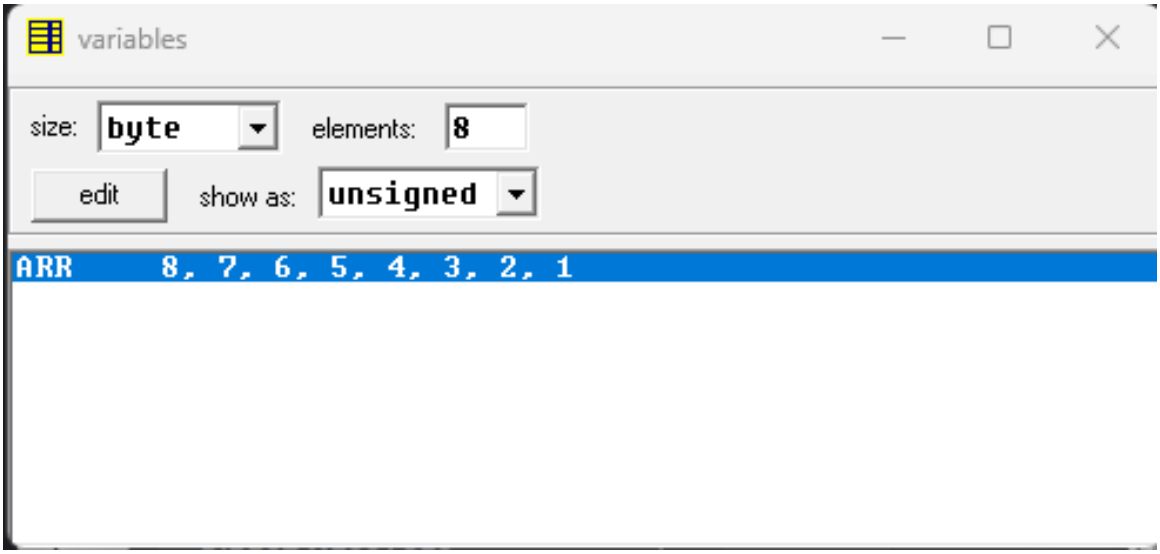


Figure 1: Reversed array.

Problem 7.2

Problem Description

Sort an array.

Process

Code:

```
    —

DATA SEGMENT
    ARR DB 7, 2, 8, 1, 0, 4, 5, 9, 6
DATA ENDS

CODE SEGMENT
    ASSUME DS:DATA, CS:CODE

START:
    MOV AX, DATA
    MOV DS, AX

    MOV BX, 8
    XOR AX, AX

    LEA SI, ARR
    MOV DX, SI

SORT_:
    MOV SI, DX
    MOV CX, BX
    MOV DI, SI
    MOV AL, [DI]

BIG_:
    INC SI
    CMP [SI], AL
    JNG NEXT
    MOV DI, SI
    MOV AL, [DI]

NEXT:
    LOOP BIG_

    MOV AL, [SI]
    XCHG AL, [DI]
    MOV [SI], AL

    DEC BX
```

```
39      JNZ SORT_
40
41  CODE ENDS
42  END START
```

Output

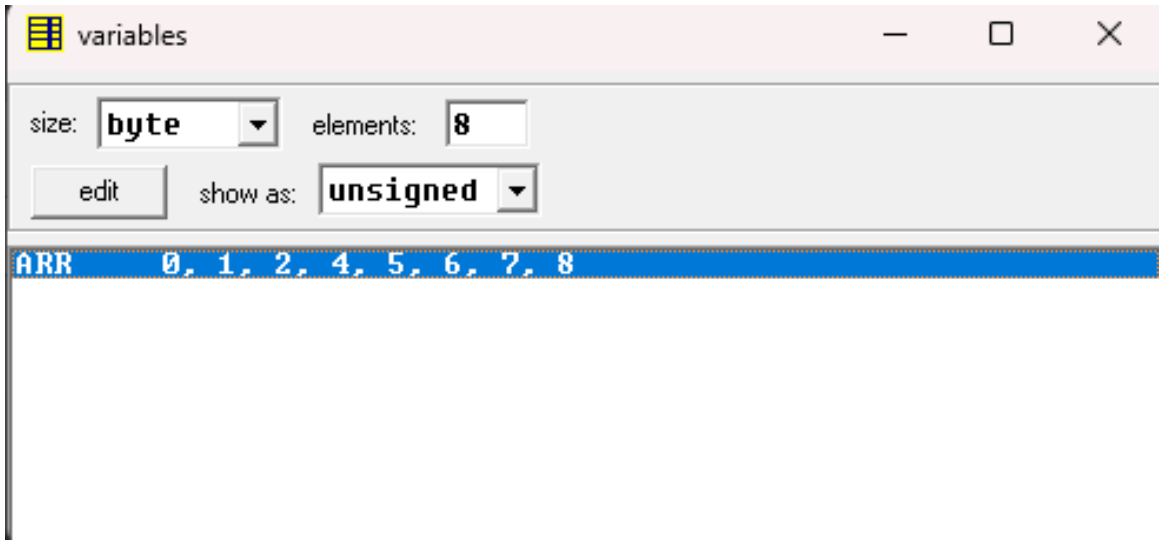


Figure 2: Sorted array.

Problem 7.3

Problem Description

Create and define two segments in memory, then exchange the contents of data segment and code segment.

Process

Code:

```
;
DATA SEGMENT
    ARR DB 1, 2, 3, 4, 5, 6, 7, 8
DATA ENDS

CODE SEGMENT
    ASSUME DS:DATA, CS:CODE

START:
    MOV AX, DATA
    MOV DS, AX

    MOV AX, 9000H
    MOV ES, AX

    MOV CX, 8
    MOV SI, 0
    MOV DI, SI

SWAP:
    MOV AL, [SI]
    XCHG ES:DI, AL
    XCHG [SI], AL

    INC SI
    INC DI
    LOOP SWAP

CODE ENDS
END START
```

Output

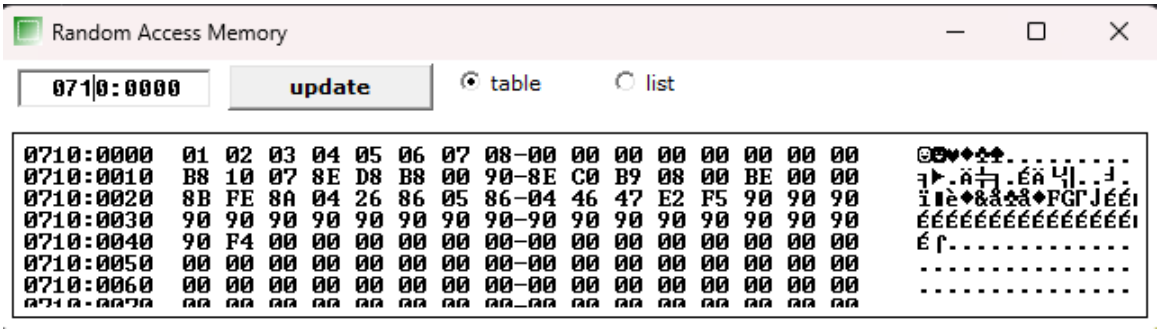


Figure 3: Code segment before swap.

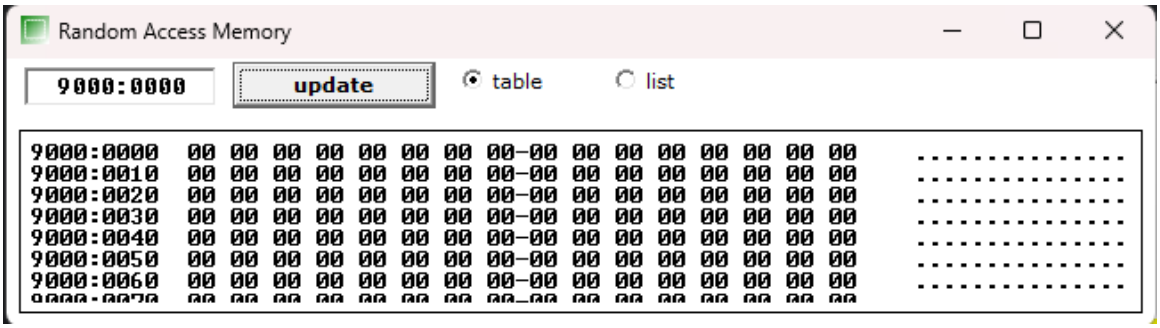


Figure 4: Extra Segment before swap.

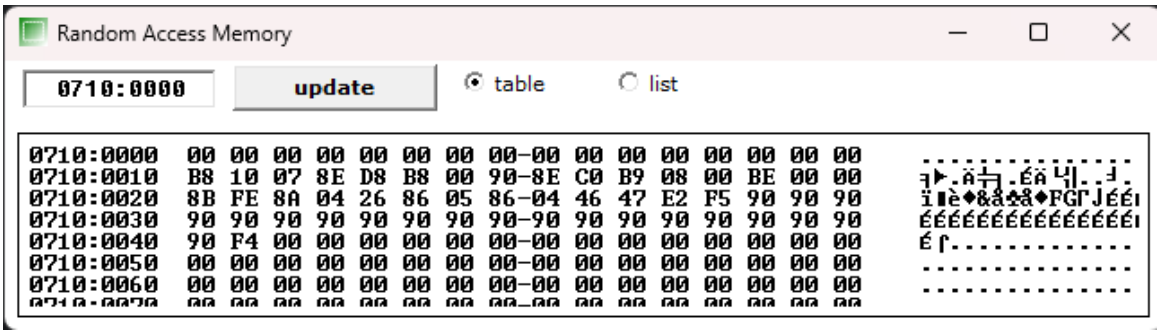


Figure 5: Code segment after swap.

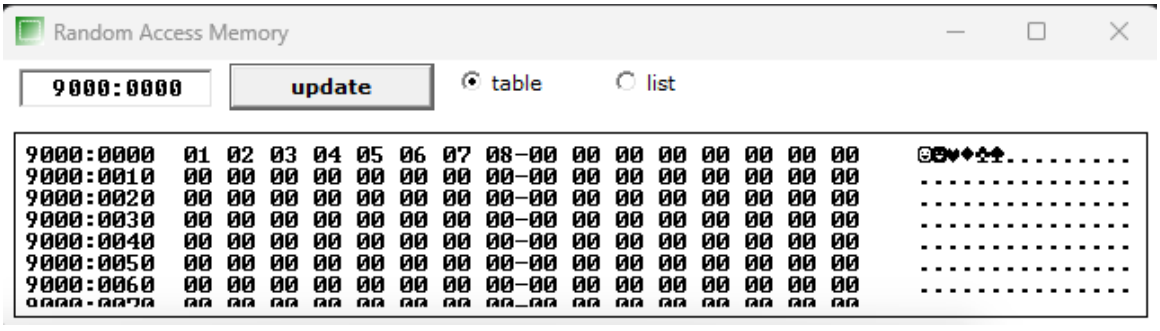


Figure 6: Extra segment after swap.

Keywords

These are the new keywords learned while solving the Problems

ARR DB 1, 2, 3, 4, 5, 6, 7, 8 This is how an array is declared. DB defines the size of each element and putting the values separated by comma defined the elements.